

## Project Planning

|                      |              |
|----------------------|--------------|
| <b>Date</b>          | 15 June 2026 |
| <b>Team ID</b>       |              |
| <b>Project Name</b>  | OptiCrop     |
| <b>Maximum Marks</b> | 5 Marks      |

### Product Backlog, Sprint Schedule, and Estimation (4 Marks)

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                            | Story Points | Priority |
|----------|-------------------------------|-------------------|------------------------------------------------------------------------------|--------------|----------|
| Sprint-1 | Data Collection               | USN-1             | As a farmer, I want to input soil parameters so the system can analyze them. | 3            | High     |
| Sprint-1 | Crop Prediction               | USN-2             | As a farmer, I want to receive crop recommendations based on soil data.      | 5            | High     |
| Sprint-2 | Analytics                     | USN-3             | As a researcher, I want to view regional crop trends.                        | 3            | Medium   |
| Sprint-2 | Policy Simulation             | USN-4             | As a policymaker, I want to simulate fertilizer impacts.                     | 4            | Medium   |

### Project Tracker, Velocity & Burndown Chart: (1 Marks)

| Sprint   | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) |
|----------|--------------------|----------|-------------------|---------------------------|
| Sprint-1 | 8                  | 5 Days   | 15 Jun 2026       | 20 Jun 2026               |
| Sprint-2 | 7                  | 5 Days   | 21 Jun 2026       | 25 Jun 2026               |

### Velocity:

Average velocity (AV) calculation:

$$AV = \frac{\text{sprint-duration}}{\text{velocity}} = \frac{15}{10} = 1.5$$